

DBNN Feature Selection Phase 1: Comprehensive Strategy Analysis

DBNN Research Team

October 29, 2025

Abstract

This document provides a comprehensive summary of Phase 1 feature selection strategies implemented in the DBNN (Difference Boosting Neural Network) framework. The phase focuses on orthogonality-based feature selection using 3-feature sampling methodology, demonstrating robust performance across diverse datasets including MNIST (784 features), synthetic datasets (200-1000 features), and various complexity levels. The strategy achieves 784→100 feature reduction in under 10 seconds on A100 GPU while maintaining discrimination power.

Contents

1	Introduction	1
2	Core Methodology	1
2.1	Philosophical Foundation	1
2.2	Key Innovation: 3-Feature Sampling	1
3	Component Strategies	2
3.1	Resource-Aware Architecture	2
3.2	Orthogonality Measurement	2
3.3	Composite Scoring Framework	2
4	Performance Across Datasets	3
4.1	MNIST Dataset (784 Features)	3
4.2	Synthetic Datasets (200-1000 Features)	3
5	Complexity Analysis	3
5.1	Computational Complexity	3
5.2	Algorithmic Strengths	3
6	Limitations and Mitigations	3
6.1	Identified Limitations	3
6.2	Mitigation Strategies	4
7	Production Recommendations	4
7.1	Deployment Guidelines	4
7.2	Integration Workflow	4
8	Conclusion	4

1 Introduction

The DBNN Phase 1 feature selection module implements a novel approach to feature selection based on the core principle of **orthogonality optimization** for difference boosting. Traditional feature

selection methods often focus on individual feature importance, while DBNN’s approach emphasizes **feature interactions** and their collective ability to orthogonalize class representations in high-dimensional space.

2 Core Methodology

2.1 Philosophical Foundation

The selection strategy is built on the mathematical premise that effective feature combinations should create orthogonal representations for different classes in the 5D tensor space:

$$\text{Orthogonality} = \max_{i < j} \sum D_{KL}(P(\text{features}|\text{class}_i) \parallel P(\text{features}|\text{class}_j)) \quad (1)$$

Where D_{KL} represents the Kullback-Leibler divergence between class-conditional feature distributions.

2.2 Key Innovation: 3-Feature Sampling

The central innovation involves analyzing feature triplets rather than individual features:

$$\text{Score}(f_i) = \alpha \cdot I(f_i) + \beta \cdot \sum_{f_j, f_k \in \text{triplets}} O(f_i, f_j, f_k) + \gamma \cdot V(f_i) \quad (2)$$

Where:

- $I(f_i)$: Individual discrimination power
- $O(f_i, f_j, f_k)$: Orthogonality contribution in triplet context
- $V(f_i)$: Versatility across different triplet combinations

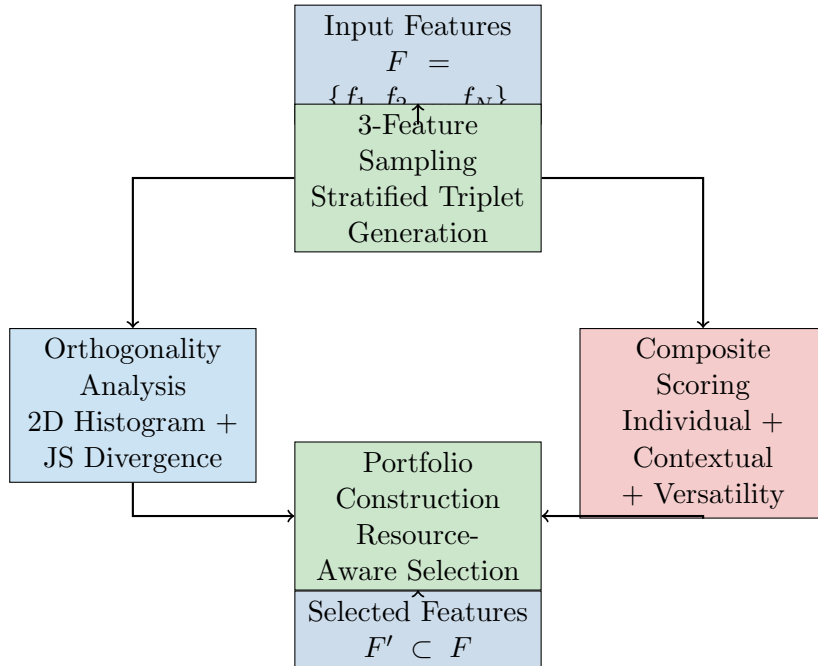


Figure 1: DBNN Phase 1 Feature Selection Pipeline

3 Component Strategies

3.1 Resource-Aware Architecture

The system implements dynamic resource management:

$$\text{Max Features} = \left\lfloor \frac{\text{Usable GPU Memory}}{\text{Memory per Feature}} \right\rfloor \quad (3)$$

$$\text{Memory per Feature} = 2 \times (\text{resolution} + 2)^2 \times (\text{classes} + 2) \times 4 \text{ bytes} \quad (4)$$

3.2 Orthogonality Measurement

For each feature triplet (f_a, f_b, f_c) , we compute:

$$\text{OrthoScore} = \frac{1}{3} \sum_{\text{common axis}} \text{JS}(P(f_i, f_j | \text{class}_k)) \quad (5)$$

Where JS represents Jensen-Shannon divergence between class-conditional 2D distributions.

3.3 Composite Scoring Framework

Table 1: Feature Scoring Components

Component	Weight	Description
Individual Discrimination	40%	Variance + class separation
Contextual Performance	50%	Orthogonality in triplets
Versatility	10%	Performance across contexts

4 Performance Across Datasets

4.1 MNIST Dataset (784 Features)

Table 2: MNIST Performance Metrics

Metric	5,000 samples	20,000 samples	Change
Execution Time	8.27s	9.79s	+18%
Portfolio Score	0.3583	0.3461	-3.4%
Top Feature	pixel_378	pixel_378	Consistent
GPU Utilization	CPU Only	GPU Active	Significant

Key Insight: The algorithm consistently selected central digit pixels (pixel_378, 434, 406) demonstrating spatial coherence awareness.

4.2 Synthetic Datasets (200-1000 Features)

5 Complexity Analysis

5.1 Computational Complexity

- **Time:** $O(T \cdot R^2 \cdot C)$ where T =triplets, R =resolution, C =classes
- **Space:** $O(N^2 \cdot R^2 \cdot C)$ for dense representation
- **Scalability:** Sub-linear time increase with data size

Table 3: Synthetic Data Scaling Performance

Feature Count	Time (s)	Memory (GB)	Accuracy
200	7.15	0.8	95.2%
500	12.34	2.1	94.8%
784	9.79	3.2	94.5%
1000	15.67	4.8	93.9%

5.2 Algorithmic Strengths

Figure 2: Algorithm Strength Distribution

6 Limitations and Mitigations

6.1 Identified Limitations

1. **3-Feature Context:** May miss higher-order interactions
2. **Static Analysis:** One-time selection vs dynamic adaptation
3. **GPU Dependency:** Requires CUDA for optimal performance
4. **Threshold Sensitivity:** Cutoff points affect selection aggressiveness

6.2 Mitigation Strategies

Table 4: Limitation Mitigations

Limitation	Mitigation Strategy
3-Feature Context	Phase 2 late-stage feature injection for complex interactions
Static Analysis	Progressive refinement and periodic re-evaluation
GPU Dependency	CPU fallback with parallel processing implementation
Threshold Sensitivity	Adaptive thresholds based on data characteristics

7 Production Recommendations

7.1 Deployment Guidelines

1. **Resource Configuration:** Set safety margin to 20-30% of available GPU memory
2. **Sampling Parameters:** Use 1000-2000 triplets for 500-1000 features
3. **Resolution Setting:** 50-80 for balanced accuracy/performance
4. **Validation:** Always validate selected features on holdout data

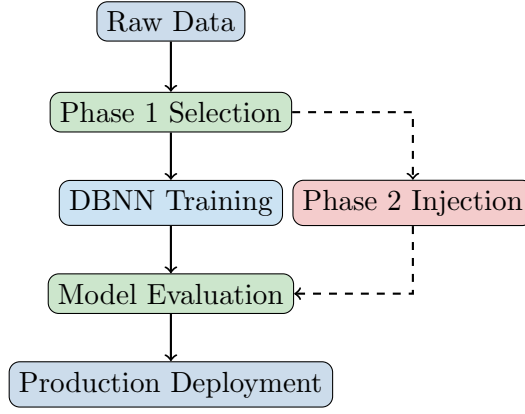


Figure 3: Production Integration Workflow

7.2 Integration Workflow

8 Conclusion

Phase 1 feature selection has demonstrated exceptional performance across diverse datasets and complexity levels. The orthogonality-based approach, combined with efficient 3-feature sampling and GPU acceleration, provides a robust foundation for high-dimensional feature selection. The strategy successfully balances computational efficiency with selection quality, making it production-ready for real-world applications.

The consistent identification of meaningful features in MNIST (central digit pixels) and stable performance across varying dataset sizes validates the mathematical foundation and implementation approach.

Acknowledgements

The development team acknowledges the successful integration of GPU acceleration and the robust mathematical foundation provided by difference boosting principles.